

Geofray Paul J

Medical image analysis · Computational radiology, neuroimaging and computational pathology

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RESEARCH INTERESTS

Failure modes of medical image analysis models and how to measure them: small-lesion recall under class imbalance, validation optimism under centre and prevalence shift, rare-class label assignment, and annotation quality as a ceiling on measured performance. Applied in computational radiology, chiefly neuroimaging (traumatic brain injury, intracranial aneurysms on CTA and MRA, post-mortem fetal brain MRI); in computational pathology (whole-slide images of brain tissue); and in endoscopy.

EDUCATION

B.Tech, Computer Science and Engineering (Artificial Intelligence)

2020 – 2024

Karunya Institute of Technology and Sciences, Coimbatore, India

PUBLICATIONS

- [1] **Geofray Paul J.** Diagnosing the Recall Tail: A Class-Imbalance Study of nnU-Net for Detection and Segmentation of Heterogeneous Moderate–Severe TBI Lesions. *AIMS-TBI Workshop, MICCAI 2026, Strasbourg, France*. Lecture Notes in Computer Science, Springer. Accepted, to appear. Sole author.

OTHER OUTPUTS

- [2] **Geofray Paul J.** Your GPU Is Lying To You: And So Is Your Test Harness, Your Docstring, Your Validation Set, and Your Cross-Validation. *MICCAI Educational Challenge 2026*, finalist. [OpenReview](#) · code: github.com/GeofrayPaulJ/MEC
- [3] **Geofray Paul J.** Below the Noise Floor: A Domain-Generalised ConvNeXt Ensemble for Barrett's Neoplasia Detection. Method report, *RARE26 challenge (EndoVis, MICCAI 2026)*, submitted. Code: github.com/GeofrayPaulJ/rare26-submission
- [4] RARE26 organisers and participating teams, including **Geofray Paul J.** Journal article summarising the RARE26 challenge (EndoVis, MICCAI 2026). In preparation; co-author as a participating team, at the organisers' invitation.

CHALLENGE RESEARCH: MICCAI 2026, SOLO ENTRIES

AIMS-TBI 2026. Lesion detection and segmentation in moderate–severe traumatic brain injury, T1-weighted MRI.

Result. 8th of 22 entries on the final detection leaderboard (balanced accuracy 0.846); 11th of 17 on segmentation (lesion Dice 0.505) (official ranking announced 1 October 2026).

Method. Self-configuring nnU-Net (ResEnc-M, 3D, five-fold ensemble) trained on 552 subjects. One model serves both tasks; the detection threshold was fixed on out-of-fold predictions and never tuned on later data.

Finding. Much of the apparent boundary error was a small-lesion recall tail: 22 of 305 cases (7.2%) failed outright, none above the median lesion volume, and excluding them lowered mean ASSD from 8.55 to 6.19 mm. Threshold lowering, batch Dice and test-time mirroring left the tail intact. Paper [1].

MICCAI Educational Challenge 2026. Finalist.

Tutorial [2]. Built from the AIMS-TBI container work: six instruments that reported success while being wrong (a CUDA availability check masking a silent CPU fallback, a test harness, a docstring, a validation set, a calibrated threshold, cross-validation), each with its mechanism and the rule that catches it, plus a ten-point preflight checklist that runs on synthetic data.

TopAneu 2026, Task 2. Vessel-specific intracranial aneurysm segmentation, CTA and MRA.

Result. 10th of 30 entries, Task 2 final test phase (mean position 10.1 across seven metrics); Dice 0.1521 against the winning entry's 0.2632. Task 1 result pending.

Method. nnU-Net v2 detection with separate CTA and MRA models, then gradient-boosted assignment of each aneurysm to one of 52 vessel sites from laterality-invariant geometric features, with vessel context from the organisers' public TopBrain model and a second classifier rejecting false positives; 218 s per CTA case (limit 720 s).

Finding. Detection was not the bottleneck. Most errors were correct detections assigned to the wrong vessel, concentrated in sites with fewer than ten training examples; nine of the 52 classes had none.

RARE26 (EndoVis). Barrett's oesophagus neoplasia classification at 1:100 prevalence, endoscopy.

Result. Open Development phase: AUROC 0.6466, PPV@90Recall 0.0113. Co-author of the challenge overview paper [4].

Method. Five-member ConvNeXt-Base ensemble with augmentation at two to three times typical magnitude against acquisition-centre shortcuts; near-duplicate-grouped and leave-one-centre-out cross-validation; a 100-image external cohort held for evaluation only.

Finding. Both internal estimates overshoot the platform result (PPV@90Recall 0.0113): pooled out-of-fold by about sevenfold, leave-one-centre-out by two- to threefold. Pooled cross-validation was the more misleading estimate under the shift between centres. The cause of the gap remains unresolved after ten tested hypotheses. Report [3].

RESEARCH ENGINEERING EXPERIENCE

Junior AI Engineer, Sudha Gopalakrishnan Brain Centre, IIT Madras

Chennai · November 2025 – present

- **Post-mortem fetal brain MRI.** nnU-Net and nnSAM skull-stripping models; found and fixed a silent orientation error that transposed predictions, and set the brain-boundary convention against a public fetal brain atlas. nnU-Net trained on masks including CSF, with fold 0 fine-tuned on expert-annotated masks.
- **Grey/white matter segmentation in multiple sclerosis.** HookNet on H&E whole-slide images (Dice 0.85 grey matter, 0.75 white matter). Traced a flat boundary metric to coarse white-matter annotations rather than the model, using a per-class surface-Dice tolerance sweep; effort was redirected to re-annotation instead of further training.
- **CD34 microvessel segmentation.** nnU-Net v2 pipeline quantifying microvessel density across gigapixel whole-slide images (Dice 0.71); removed tile-boundary artefacts with Gaussian-weighted overlap-averaged inference.
- **IHC analysis platform.** Multi-marker inference (CD68, CD34, astrocytes) behind one React, Django and Celery/Redis service on a DGX A100; resolved container crashes on TIFF workloads of up to 150 GB.
- **Specimen provenance.** Mapped the brain-specimen lifecycle (acquisition, sectioning, staining, QR-coded tracking, scanning, storage) and its physical edge cases through interviews with laboratory staff; designed the relational data model, built the tracking interface, and delivered BPMN/UML provenance diagrams to the CSO.
- **Infrastructure.** Persistent, containerised GPU training on the DGX A100 (Docker, tmux). Also built the centre's public website (sgbc.humanbrain.in) and the BRICS 2026 Neuroscience Symposium pages.

Career break (health), June 2024 – October 2025. Fully recovered.

ONGOING WORK

ARBITER, a benchmark for vision-language models reading charts. Tasks are generated by code, so every answer is exact; difficulty changes one factor at a time, and a search finds where each model starts to fail (Qwen3-VL, Gemma 3, InternVL3.5, MiniCPM-V). Synthetic by design: medical labels vary between annotators and cannot be shared publicly, so they cannot validate a measuring instrument.

METHODS AND TOOLS

Models: nnU-Net v2 (ResEnc presets), HookNet, ConvNeXt ensembles, gradient-boosted trees; SAM, MedSAM, nnSAM, PathoSAM, UNI, CONCH; LoRA fine-tuning.

Evaluation: near-duplicate-grouped and leave-one-centre-out cross-validation, external cohorts, bootstrap confidence intervals, thresholds fixed before test; Dice, surface Dice, HD95, ASSD, AUROC, PPV at fixed recall, MCC.

Imaging data: structural and post-mortem fetal MRI, CTA and MRA; gigapixel whole-slide images (H&E, IHC); endoscopy; DICOM, NIfTI; MRI preprocessing, orientation and annotation quality control.

Compute: PyTorch, CUDA, AMP and bf16, DDP, Nsight Systems; Docker and time-limited containerised inference on grand-challenge.org; checksum-based provenance; Linux, tmux; DGX A100.

Software: Python, TypeScript, SQL; Django REST Framework, Celery, Redis; React, Next.js; vLLM serving.

LANGUAGES

English (fluent), Tamil (native), Spanish (A2).